MUHAMMAD SHERAZ

Plant Molecular Biology, Genomics and Bioinformatics
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RESEARCH PROFILE

Plant molecular biologist and computational genomicist working on transcription factor gene family characterisation and stress-responsive gene regulation in crop and woody perennial species. First author of a peer-reviewed genome-wide study of the GLK transcription factor family in Persian walnut, published in Integrative Plant Biotechnology and presented additionally as a conference abstract. Author of a second independent genome-wide study of the BBR-BPC transcription factor family in soybean under salinity stress, combining hidden Markov model profiling, comparative synteny and Ka/Ks selection analysis, genome-wide motif-regulon prediction, molecular docking, and RNA-seq differential expression across three public BioProjects, with pipelines built in Python on Linux. Research interests centre on how transcription factor families evolve by duplication and how their regulatory targets mediate abiotic and biotic stress tolerance — questions directly relevant to crop improvement for marginal and saline environments.

EDUCATION

BSc (Hons.) Agricultural Biotechnology | September 2021 – May 2025
University of Agriculture Faisalabad (UAF), Pakistan

- Final grade: 3.89 / 4.00 CGPA (82.24% aggregate). Departmental topper.
- Awarded a full merit scholarship covering tuition for the entire degree.
- Core coursework: Recombinant DNA Technology I and II, Plant Transformation, Molecular Cell Biology, Bioinformatics and Functional Genomics, Fundamentals of Seed Molecular Biology, Diagnostic Plant Pathology, Nanobiotechnology, Virology, Biostatistics.

Bachelor of Education (BEd) | February 2024 – October 2025 — *University of Agriculture Faisalabad (UAF), Pakistan — Final grade: 3.33 / 4.00 CGPA*

Intermediate, FSc (Pre-Medical) | 2018 – 2020 — *Punjab Group of Colleges, Faisalabad, Pakistan — 1021 / 1100 marks, Grade A+*

PEER-REVIEWED PUBLICATIONS

1. **Sheraz, M.**, Ali, A., and Awan, F. S. (2026). Phylogenetic classification, characterization and expression of Golden2-like (GLK) transcription factor gene family in Persian walnut in response to *Phomopsis capsici* infection. *Integrative Plant Biotechnology*, 4(3), 249–266. DOI: 10.55627/pbiotech.004.03.2010. Published online 28 September 2026. Open access (CC BY 4.0). **First author.**

   Abstract. The G2-like (GLK) transcription factors are members of the GARP superfamily and control chloroplast biogenesis, photosynthesis development and stress responses in plants. However, they are yet to be characterised in a systematic manner in woody perennial species. In this work, we identified and characterised the GLK gene family at genome level in Persian walnut (Juglans regia L.) using the chromosome-scale Walnut 2.0 genome assembly (GCF_001411555.2). Forty-four non-redundant JrGLK genes were identified at the family level; of these, 43 with complete genomic coordinates were mapped and found unevenly distributed on 14 of the 16 walnut chromosomes, with the largest gene clusters on chromosomes 3 and 4. A total of 126 GLK protein sequences from J. regia, A. thaliana and O. sativa were subjected to maximum-likelihood phylogenetic analysis, which classified the family into 15 statistically supported groups. Members from A. thaliana and O. sativa were distributed across most groups, with no purely walnut-specific group reaching the support threshold, arguing against walnut-specific lineage expansion. Conserved motif and exon–intron structure analysis showed patterns concordant with this classification, with several groups distinguished by group-specific motif combinations. Promoter analysis of the 43 characterised genes revealed predicted occurrences of abscisic acid-, drought-, defense- and light-associated regulatory elements across the family, representing predicted motif occurrences rather than confirmed regulatory function. Synteny analysis identified 23 JrGLK genes with confirmed syntenic orthologs in A. thaliana and 23 in Populus trichocarpa, indicating evolutionary retention of GLK loci across dicotyledonous lineages. Transcriptomic analysis of plants infected with Phomopsis capsici (BioProject PRJNA867174) identified 13 significantly differentially expressed JrGLK genes, six up-regulated and seven down-regulated, indicating a transcriptional response to infection. Together, these data provide a genomic resource for the GLK family in J. regia and a foundation for functional studies and molecular breeding in walnut and other Juglandaceae species.

   Keywords. Juglans regia; genome-wide identification; G2-like transcription factors; phylogeny; abiotic stress; biotic stress.

2. Ali, A., Habib, K., **Sheraz, M.**, Fatima, A., Asif, A., Tariq, A., Tariq, I., Nawaz, A., Durrani, F., and Sattar, Z. (2026). CRISPR–Nanoparticle delivery systems for precision gene editing in plant pathogens. *Multidisciplinary Surgical Research Annals*, 4(1), 20–32. Published 19 January 2026. Third author of ten.

   Summary. Experimental study of a nanoparticle-assisted CRISPR-Cas delivery system for editing the fungal pathogen Fusarium oxysporum: polymeric nanoparticles with pH-dependent release enhanced uptake and nuclear entry, drove deletion of a virulence gene and reduced fungal growth and virulence, with high off-target specificity and no host-genome indels. Keywords: CRISPR–Cas; nanoparticle delivery; plant pathogen gene editing.

CONFERENCE ABSTRACTS AND PRESENTATIONS

1. **Sheraz, M.**, Ali, A., and Awan, F. S. (2026). Genome-wide identification, phylogenetic classification and *in silico* characterisation of the G2-like (GLK) transcription factor gene family in Persian walnut (*Juglans regia* L.). Published abstract, Plant Biotechnology Symposium, March 2026.

   Summary. Conference presentation of the genome-wide GLK transcription factor study subsequently published in Integrative Plant Biotechnology (see Peer-Reviewed Publications, entry 1). Covered family-level identification of 44 non-redundant JrGLK genes on the Walnut 2.0 assembly, maximum-likelihood phylogenetic classification into 15 supported groups across three species, conserved motif and gene-structure analysis, cross-species synteny with Arabidopsis thaliana and Populus trichocarpa, and differential expression of 13 JrGLK genes under Phomopsis capsici infection.

2. **Sheraz, M.** and Ali, A. (2026). Genome-wide identification and characterization of the GmBBR-BPC transcription factor family in soybean (*Glycine max*): evolution, regulon architecture and transcriptional responses to salt stress. Published abstract, 2nd Congress of the Biotechnology Society of Pakistan (BSP) on Biotechnology for Global Health and Food Security, University of Agriculture Faisalabad, 27–28 October 2026. **Manuscript in preparation; wet-laboratory validation ongoing.**

   Abstract. BARLEY B RECOMBINANT/BASIC PENTACYSTEINE (BBR/BPC) proteins are plant-specific GAGA-binding transcription factors that regulate development and stress responses, but the family has not been characterised genome-wide in soybean (Glycine max). Here we identified ten non-redundant GmBBR-BPC genes on chromosomes 4–9 of the Wm82.a4.v1 genome using three independent searches (a Pfam PF06217 profile, an Arabidopsis-derived profile and reciprocal BLASTP), which converged on the same ten loci. All encoded proteins carry the conserved GAGA-binding domain. Phylogenetic analysis assigned them to Class I (GmBBR-BPC2, 5, 6, 8, 9, 10) and Class II (GmBBR-BPC1, 3, 4, 7). Four whole-genome-duplication-derived paralog pairs (BPC5/9, BPC2/8, BPC1/4, BPC3/7) were identified, and Ka/Ks ratios of 0.052–0.192 indicate strong purifying selection. Promoter analysis revealed abundant stress-, hormone- and light-responsive cis-elements. A genome-wide scan of 56,044 genes showed that GA-repeat and C-box motifs co-occur more often than expected (odds ratio 3.20), and the resulting regulon was enriched for seven GO terms. To examine salt responses, we re-analysed three public RNA-seq datasets from raw reads with one uniform pipeline. In a replicated leaf time course, eight of 60 gene-by-contrast tests were significant after family-wide correction, all at 24 or 48 h. The changes were modest (|log2 fold change| up to 0.99, excluding the low-abundance GmBBR-BPC7) and did not differ between a tolerant and a sensitive cultivar. Only GmBBR-BPC1 decreased significantly, in the same direction, in two independent leaf experiments, whereas the direction of GmBBR-BPC2, GmBBR-BPC7 and GmBBR-BPC8 depended on the experiment. Results were robust to multi-mapping reads, to the statistical model and to fold-change shrinkage. Together, these data indicate that the GmBBR-BPC family is a modest, context-dependent component of the soybean salt response and provide a basis for functional studies.

   Keywords. Glycine max; BBR/BPC; GAGA-binding factor; gene family; salt stress; RNA-seq; cis-elements; regulon.

RESEARCH EXPERIENCE

Independent Research — GmBBR-BPC Transcription Factor Family in Soybean | 2025 – present
Conducted at the Centre of Agricultural Biochemistry and Biotechnology, University of Agriculture Faisalabad, Pakistan. Published conference abstract; manuscript in preparation.

- Identified and validated 10 non-redundant BBR-BPC transcription factor loci across the 74,206-protein soybean reference proteome using three independent methods — Pfam PF06217 hidden Markov model profiling (HMMER / pyhmmer), a custom-built HMM, and reciprocal BLASTP — all converging on an identical gene set, with a 306-bit score separation from the nearest non-member.
- Constructed maximum-likelihood phylogenies (MEGA, ClustalW, MUSCLE) and resolved the family into two structural classes; confirmed GAGA-binding domain architecture across all members via NCBI Conserved Domain Database.

- Performed interspecies synteny and Ka/Ks selection analysis against Arabidopsis thaliana, rice and tomato using MCScanX with Nei–Gojobori Z-testing, confirming strong purifying selection; identified four whole-genome-duplication-derived segmental paralog pairs.
- Developed Python pipelines to scan 56,044 gene promoters genome-wide for GAGA regulatory motifs, producing a tiered candidate regulon with significant element co-occurrence statistics and gene ontology enrichment cross-validated across three independent platforms.
- Executed a complete molecular docking workflow in HDOCK: blind docking, randomised negative controls, motif-restrained redocking, self-dimerisation analysis, AlphaFold pLDDT confidence assessment, and hypergeometric testing of DNA-binding and dimerisation interface overlap.
- Processed RNA-seq datasets from three public NCBI BioProjects using fastp, FastQC, HISAT2, featureCounts and DESeq2 / pydeseq2, running pipelines on Galaxy and locally under Linux.
- Re-analysed three public soybean RNA-seq salt-stress datasets from raw reads under one uniform pipeline, applying family-wide multiple-testing correction and testing robustness to multi-mapping reads, statistical model choice and fold-change shrinkage; reported the family as a modest, context-dependent component of the salt response rather than overstating effect size.
- Designed and validated quantitative real-time PCR primers (NCBI Primer-BLAST, IDT OligoAnalyzer) with complete secondary-structure documentation; wet-laboratory salinity treatment and expression validation currently in progress.

Independent Research — JrGLK Transcription Factor Family in Persian Walnut | 2024 – 2026
Conducted at the University of Agriculture Faisalabad, Pakistan. Resulted in a first-author peer-reviewed publication and a conference abstract.

- Identified and characterised 44 non-redundant GLK transcription factor genes on the Persian walnut Walnut 2.0 chromosome-scale assembly, mapping 43 across 14 of 16 chromosomes with gene structure and conserved motif analysis (MEME Suite).
- Built a maximum-likelihood phylogeny of 126 proteins across three species using IQ-TREE 2 with ModelFinder selection and 1,000 ultrafast bootstrap replicates, resolving 15 statistically supported groups.
- Conducted MCScanX synteny analysis identifying 23 syntenic orthologs in Arabidopsis thaliana and 23 in Populus trichocarpa, visualised with Circos.
- Analysed RNA-seq expression under Phomopsis capsici fungal infection (BioProject PRJNA867174) using Bowtie2 alignment, featureCounts quantification and DESeq2, identifying 13 significantly differentially expressed genes.
- Wrote a custom strand-aware Python script for promoter extraction with neighbouring-gene boundary trimming; mapped cis-regulatory elements via PlantCARE.
- Managed the complete peer-review cycle, including a point-by-point response to two reviewers and galley proof correction.

Research Intern | September 2024 – September 2025 — *State Research Institute Centre for Physical Sciences and Technology (FTMC), Vilnius, Lithuania*

- Completed a competitively awarded 12-month international research placement applying molecular and analytical techniques in a multidisciplinary physical sciences research environment.
- Worked within a multilingual research team, delivering technical communication across institutional and cultural boundaries.

Research Intern | February 2025 – April 2025 — *Agricultural Biotechnology Research Institute (ABRI), Ayub Agricultural Research Institute, Faisalabad, Pakistan*

- Supported laboratory experimentation, data collection and documentation for ongoing biotechnology research projects.

Research Intern | June 2024 – September 2024 — *Vegetable Research Institute, Ayub Agricultural Research Institute, Faisalabad, Pakistan*

- Gained applied research experience in plant pathology and crop science within a national agricultural research institute.

TECHNICAL SKILLS

Comparative and Evolutionary Genomics: HMMER, pyhmmer, BLAST / BLASTP, MCScanX, Circos, Ka/Ks and Nei–Gojobori selection analysis, synteny and collinearity analysis, whole-genome duplication analysis, PlantTFDB, Ensembl Plants, NCBI RefSeq

Phylogenetics: IQ-TREE 2, MEGA, ModelFinder, ClustalW, MUSCLE, trimAl, ultrafast bootstrap assessment, iTOL

Transcriptomics: RNA-seq analysis, fastp, FastQC, MultiQC, Bowtie2, HISAT2, featureCounts, DESeq2, pydeseq2, differential expression analysis, count matrix construction, Galaxy

Regulatory and Motif Analysis: MEME Suite, PlantCARE, NCBI Conserved Domain Database, custom promoter extraction, genome-wide motif scanning, gene ontology enrichment (ShinyGO, g:Profiler)

Structural and Network Biology: HDOCK molecular docking, AlphaFold and pLDDT confidence assessment, STRING, Cytoscape, protein–protein interaction networks

Programming and Computing: Python (pandas, NumPy, SciPy, BioPython, custom pipeline development), Linux / Ubuntu, Bash scripting, Conda, Git, Google Colab, Jupyter, TBtools, MapChart

Statistics: Hypergeometric testing, Wilcoxon rank-sum test, false discovery rate correction, odds-ratio and co-occurrence analysis, experimental design

Laboratory and Molecular Biology: Recombinant DNA technology, molecular cloning, plant tissue culture and transformation, PCR, quantitative real-time PCR, primer design and validation, DNA and RNA extraction, gel electrophoresis, hydroponic stress treatment design, aseptic technique

Scientific Communication: Peer-reviewed manuscript preparation, peer-review response, grant proposal writing, conference abstract preparation, scientific presentation

PROFESSIONAL SERVICE AND LEADERSHIP

Organising Committee Member — 2nd BSP Congress on Biotechnology for Global Health and Food Security | October 2026 — *Biotechnology Society of Pakistan, hosted at University of Agriculture Faisalabad, Pakistan*

- Managed abstract submission, review and compilation for a national scientific congress.
- Designed and automated the registration and confirmation system, and handled delegate correspondence and logistics.

Teaching Intern | February 2023 – May 2023 — *WAPDA Boys High School, Faisalabad, Pakistan — Delivered classroom instruction and supported academic supervision and record keeping.*

Volunteer and Presenter — Public Health Awareness Seminar and Walk | December 2024 — *University of Agriculture Faisalabad — Presented technical health information to a general public audience and led the accompanying outreach event.*

Event Coordinator (Volunteer) — National Championship Events | February – March 2024 — *University of Agriculture Faisalabad — Coordinated logistics and on-site operations for two live-broadcast public events.*

AWARDS AND HONOURS

- Departmental Topper, BSc (Hons.) Agricultural Biotechnology, University of Agriculture Faisalabad
- Merit Scholarship, University of Agriculture Faisalabad — full tuition remission across the BSc degree
- Competitively Selected International Research Intern, State Research Institute Centre for Physical Sciences and Technology, Vilnius, Lithuania (12-month placement)
- Merit-Based Awardee, Prime Minister's Laptop Scheme Phase 3, Government of Pakistan
- Selected Participant, Chief Minister Punjab Agricultural Graduate Internship Scheme

CERTIFICATIONS AND TRAINING

- Workshop: Molecular Biology in Forensic Sciences, University of Agriculture Faisalabad (2024)
- Workshop: Computer-Aided Development of Antimicrobial Agents, University of Agriculture Faisalabad (2023)
- Workshop: One-Week Hands-On Laboratory Training, University of Agriculture Faisalabad (2023)
- Workshop: Microsoft Office, University of Agriculture Faisalabad (2023)

LANGUAGES

English — full professional proficiency (CEFR C1). Urdu — native. Punjabi — native (CEFR C2).

REFERENCES

Prof. Dr. Faisal Saeed Awan — Professor, Centre of Agricultural Biochemistry and Biotechnology, University of Agriculture Faisalabad. Email: faisal.saeed@uaf.edu.pk. ORCID: 0000-0003-1525-8373. (Research supervisor and corresponding author, publication 1.)